

Research Design: A Multi-Modal XAI Deep Learning framework for pneumonia diagnosis: Evaluating diagnostic performance and clinical trust over diverse populations.

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Literature Review is submitted in partial fulfilment of the requirements for the Bachelor
of

Computing Degree

at Belgium Campus iTversity

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Date of submission: 30/08/2026

1. Introduction:

Computer-Aided Diagnosis (CAD) systems have been transformed by the integration of Artificial Intelligence (AI), enabling deep learning models to achieve higher levels of diagnostic accuracy in the automated detection of pneumonia from chest X-ray (CXR) images. The diagnostic performance of the CAD deep learning models has also seen significant improvements due to recent advancements in Deep Convolution Neural Networks (DCNNs), transfer learning, and multi-modal learning. Furthermore, the integration of Explainable AI (XAI) and Human-in-the-Loop (HITL) strategies provides greater support for transparency and clinical involvement. These developments demonstrate the potential of AI-assisted diagnostic systems to enhance clinical decision-making by reducing the burden on healthcare professionals.

Despite these enhancements, the previous literature review paper identified several limitations that continue to prevent the widespread adoption of AI-assisted pneumonia diagnostic systems. These limitations indicate that existing approaches address diagnostic accuracy, interpretability, generalisability and clinician-centred decision support as separate research problems rather than integrating them into a unified framework. Furthermore, many proposed solutions achieve improvements in one of these areas while creating limitations in another, limiting the reliable deployment of diagnostic systems across diverse healthcare environments. This study addresses the research gap by proposing the design of a **multi-modal Explainable Artificial Intelligence Deep Learning Framework for Pneumonia Diagnosis**. This framework integrates transfer-learning, multi-modal data fusion, XAI techniques and HITL decision support within a single diagnostic architecture that seeks to balance diagnostic accuracy with transparency, robustness and clinical trust to better support real-world healthcare applications.

The primary objective of this research is not only to analyse existing diagnostic approaches, but to design, develop, and evaluate an integrated diagnostic framework capable of addressing these identified limitations. The primary contribution of this research study is the design and evaluation of this novel multi-modal XAI diagnostic framework. To achieve this, a Design Science Research (DSR) methodology is adopted which will provide a structured and iterative approach for developing and evaluating an innovative diagnostic framework that addresses identified real-world problems while contributing new knowledge. This DSR methodology will follow the Design Science Research Process Model proposed by Peffers, which provides a systematic process for identifying the problem, defining solution objectives, designing and developing proposed frameworks.

This study justifies the selection of the DSR research design, describing the adopted DSR process and presenting the proposed diagnostic framework with its research design decisions that highlight its development. Measures will also be implemented to ensure the methodological rigor, reliability, validity, and ethical compliance throughout the research process.

2. Justification of the Research Design:

2.1 Nature of the Research Problem:

The previous literature review identified significant advancements in deep learning, transfer learning, multi-modal learning, XAI and HITL approaches for AI-assisted CAD systems for pneumonia detection. However, despite these advancements, several limitations continue to hinder the reliable adoption of these systems in clinical practice. These limitations remain because existing studies primarily address diagnostic accuracy, interpretability, generalisability, and clinician involvement as separate research problems, preventing the development of a comprehensive diagnostic solution. Therefore, the research problem extends beyond evaluating existing approaches, moving towards designing an integrated framework that combines these AI approaches and technological areas in a complementary manner that addresses these identified limitations and challenges simultaneously.

2.2 Why Design Science Research is Appropriate:

The absence of an integrated diagnostic framework makes the Design Science Research (DSR) an appropriate methodology for this study. DSR focuses on the design, development, and evaluation of innovative artefacts and design concepts that solve real-world problems while contributing new knowledge. Unlike traditional research approaches like Systematic Literature Reviews (SLR) that synthesises and evaluates existing knowledge or experiments that evaluates hypotheses or intervention under controlled conditions, DSR is solution-oriented in which artefacts are designed, developed and evaluated. The primary DSR artefact developed in this research study is a multi-modal XAI deep learning framework for pneumonia diagnosis. This framework integrates transfer learning and multi-modal data fusion to improve diagnostic performance and prediction accuracy, while incorporating XAI techniques and HITL decision support to enhance the transparency and clinical usability of the predictions, resulting in a unified diagnostic architecture.

Peppers DSR model provides a structured process for identifying problems and defining solution objectives, leading to the design and development of artefacts that demonstrates their application, allowing for the evaluation of their effectiveness and research contributions. Similarly, the rigorous construction and evaluation of these innovative artefacts generate knowledge that indirectly contributes to future studies which closely aligns with the objectives of this research.

2.3 Alignment of Design Science Research with this Study:

The objectives of this research study seek to develop an integrated diagnostic framework that addresses the remaining limitations identified in the literature review, rather than incrementally improving an existing diagnostic model. This approach aligns with the principles of DSR methodology, in which the methodology's iterative nature supports the continuous refinement and evaluation of the proposed novel

framework throughout its development process. Its rigorous process guides the design, development, and experimental evaluation of the artefacts against established performance criteria, enabling the systematic improvements of the proposed diagnostic framework.

3. Design Science Research Process:

This study follows Peffers DSR methodology which addresses the research problem by identifying the problem, defining solution objectives, and designing and developing an artifact to demonstrate, evaluate, and communicate research results. The artifact is a lightweight multi-modal XAI deep learning framework designed for pneumonia diagnosis using CXR images and patient metadata. This framework combines DenseNet-121 transfer learning, multi-modal data integration via metadata, XAI using GRAD-CAM, HITL review, subgroup generalisability analysis, and preprocessing comparison.

The implementation of this artifact is structured into the following six experiments which were designed to evaluate the different aspects of the proposed artefact:

- **E1:** Image-only diagnostic baseline.
- **E2:** Multi-modal diagnostic model using CXR images and patient metadata.
- **E3:** Explainability using Grad-CAM.
- **E4:** Human-in-the-Loop review.
- **E5:** Generalisability across demographic and imaging subgroups.
- **E6:** Effect of preprocessing choices on diagnostic performance.

3.1 Problem Identification and Motivation:

Due to overlapping radiographic manifestations and variations across patients and imaging environments, pneumonia diagnosis from CXR images remains challenging to this day. Although deep learning has demonstrated strong automated pneumonia detection potential, existing literature indicates its predictive performance alone is insufficient for reliable pneumonia detection. Image-only models can overlook relevant patient information while other model types learn data specific or irrelevant visual features, reducing generalisability and reliability across different populations and imaging conditions. This influences the interpretability of deep learning predictions, making it difficult for clinicians to understand and evaluate automated conditions.

These limitations demonstrate the need for comprehensive diagnostic approaches that considers multiple sources of clinical information, providing interpretable predictions and model reliability evaluations across different patient subgroups. Therefore, this paper aims to design a unified framework artifact that integrates

patient metadata, visual explanation techniques, human review, subgroup evaluation, and appropriate image preprocessing within the diagnostic process.

3.2 Objectives of the Solution:

The primary objective is to design and evaluate a lightweight multi-modal diagnostic framework that improves the predictive performance, interpretability, human involvement and generalizability of pneumonia classification. The framework will utilize **DenseNet-121 transfer learning** as an image-based baseline and extend it by integrating patient metadata to determine if multi-modal information improves diagnostic performance. **Grad-CAM** is then incorporated to provide visual explanations of model predictions across true-positive/negative and false-positive/negative cases. A HITL mechanism is also integrated to capture reviewer agreement and disagreement, enabling the investigation of human interaction with model predictions. The framework also evaluates performance across age, sex, and CXR view subgroups to assess potential differences in diagnostic behaviour while investigating the effect of **CLAHE** as an alternative preprocessing strategy.

Together, these components will enable the framework to be evaluated across predictive performance, explainability, human correction, subgroup behaviour, and preprocessing sensitivity, extending the solution beyond a standalone pneumonia classifier toward an integrated diagnostic decision-support artefact.

3.3 Design and Development of the Artefact:

The developed artefact is a DSR research prototype designed to investigate a multi-modal approach to binary pneumonia classification using the NIH ChestX-ray14 dataset. The framework is structured as a sequential pipeline consisting of **data preparation, image classification, metadata fusion, explainability, human review, and evaluation components**. The design progressively extends an image-only baseline into a multi-modal diagnostic decision-support framework and subsequently evaluates its behaviour under different conditions.

- **Data Preparation:** Prepares the NIH ChestX-ray14 dataset for binary pneumonia classification by retaining cases labelled Pneumonia and No Finding. Patient metadata consisting of age, sex, and radiographic view position is encoded alongside the corresponding CXR images. To prevent images from the same patient from occurring across different data partitions, the dataset is divided at patient level into 70% training, 15% validation, and 15% test sets, using a fixed random seed. This produces a held-out test set for subsequent evaluation while reducing the risk of patient-level data leakage.
- **E1 - Image-only Baseline:** Establishes the baseline diagnostic architecture using an ImageNet-pretrained DenseNet-121 model with CXR images as the sole input. The architecture uses the DenseNet-121 backbone to extract hierarchical image features, followed by a task-specific projection and

classification head for binary pneumonia prediction. This experiment establishes the reference architecture against which the contribution of additional patient metadata can subsequently be assessed.

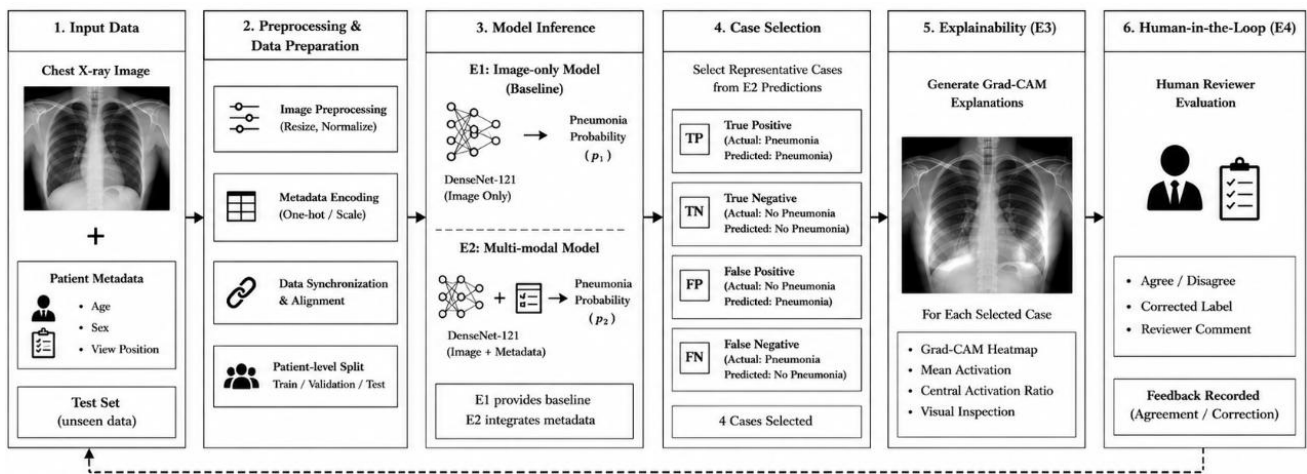
- **E2 - Multi-modal Diagnostic Model:** Extends the E1 image-only architecture by incorporating structured patient metadata consisting of age, sex, and radiographic view position. The CXR image is processed through the DenseNet-121 image encoder, while the metadata is processed through a separate metadata encoder. The resulting image and metadata representations are then fused before being passed to the final classification layer, allowing information from both visual and structured clinical sources to contribute to the pneumonia prediction.
- **E3 - Explainable Artificial Intelligence:** Extends the multi-modal model with an explainability component using Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM is applied to the final convolutional layer of the DenseNet-121 backbone to generate spatial heatmaps highlighting image regions associated with the model's prediction. The component is designed to support examination of model behaviour across both correct and incorrect predictions by providing a visual representation of the image features contributing to the model output.
- **E4 - Human-in-the-Loop:** Incorporates human oversight into the diagnostic workflow by allowing a reviewer to assess the model's predictions together with their corresponding Grad-CAM explanations. The reviewer can record agreement or disagreement with the prediction, provide a corrected label where necessary, and optionally provide a supporting comment. This component positions the model as a decision-support system in which human reviewers retain the ability to challenge and correct automated predictions.
- **E5 - Generalisability and Subgroup Evaluation:** Evaluates whether the multi-modal model maintains consistent behaviour across different patient and imaging subgroups within the held-out NIH ChestX-ray14 test set. The analysis considers age, sex, and radiographic view, with the test population divided into predefined demographic and imaging groups. Diagnostic performance is subsequently assessed within each subgroup to identify differences in model behaviour. This represents within-dataset subgroup generalisability rather than external validation, as all evaluations are performed using the NIH ChestX-ray14 test set.
- **E6 - Preprocessing Comparison:** Extends the evaluation of the multi-modal model by investigating whether the image preprocessing strategy influences diagnostic performance. Two configurations are compared: the baseline preprocessing pipeline and a CLAHE-based preprocessing pipeline. The same multi-modal architecture, dataset partitions, training configuration, and experimental conditions are maintained between the two configurations, allowing the preprocessing strategy to serve as the primary experimental variable.

The E1–E4 components form the core progressive diagnostic artefact, moving from image-only classification to multi-modal prediction, visual explanation, and human oversight. E5 and E6 constitute

additional evaluation experiments applied to the developed artefact, examining subgroup behaviour and preprocessing sensitivity respectively. Collectively, these components allow the research to investigate pneumonia classification not only in terms of predictive performance, but also in relation to multi-modal information integration, interpretability, human oversight, subgroup behaviour, and preprocessing sensitivity.

3.4 Demonstration:

The artefact was demonstrated using an end-to-end experimental workflow built on the NIH ChestX-ray14 dataset which operationalises the multi-modal diagnostic framework. It progresses from image and patient metadata preparation to model inference, representative case selection, explainability, and human review. The held-out test set was treated as unseen data, ensuring that the demonstration reflected the application of the developed artefact to previously unused cases.



Following patient-level dataset partitioning, chest X-ray images were resized and normalised, while the available patient metadata, comprising age, sex, and view position, was encoded into a numerical representation. The image and metadata inputs were synchronised using the dataset manifest before being supplied to the respective models. E1 implemented the image-only DenseNet-121 baseline, while E2 extended the architecture by combining the DenseNet-121 image representation with the encoded metadata. Both configurations were trained using the defined training set and evaluated against the validation set during training. The E2 checkpoint selected through validation performance was subsequently loaded for inference on the held-out test split.

For test-set inference, the E2 evaluation procedure loaded the trained model in evaluation mode and processed the test images without augmentation. Each image was passed through the model together with its corresponding metadata to produce a sigmoid-derived pneumonia probability. The configured classification threshold was then applied to convert each probability into a binary prediction. The resulting predictions, probabilities, ground-truth labels, and associated image paths provided the basis for subsequent evaluation and representative case selection. Four representative cases were selected according to their prediction outcomes: a true positive (TP), true negative (TN), false positive (FP), and false negative (FN). This

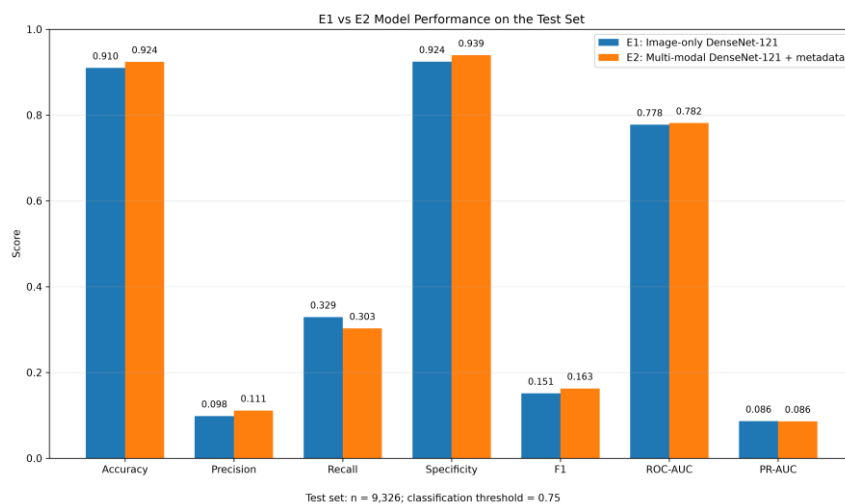
selection ensured that both correct and incorrect diagnostic behaviours were included in the subsequent demonstration of the artefact. The selected cases were then processed through E3 to demonstrate the explainability component. Grad-CAM was generated from the model to produce a visual explanation of the image regions contributing to each prediction. For each case, the resulting Grad-CAM heatmap and overlay were retained for visual inspection, together with quantitative measures including mean activation and central activation ratio. These outputs provided an interpretable representation of the model's decision-making process for the selected diagnostic cases.

Finally, E4 demonstrated the human-in-the-loop component using the generated cases and explanations. The implemented review interface, available through both a command-line workflow and Streamlit application, presented the case image, model probability, predicted class, and corresponding Grad-CAM overlay to the reviewer. For each case, the reviewer recorded whether they agreed or disagreed with the model prediction, supplied a corrected diagnostic label, and optionally provided a qualitative comment. The resulting feedback was stored for subsequent HITL analysis. The demonstration was therefore completed as a sequential workflow in which model predictions were generated, representative cases were explained, and the resulting predictions and explanations were subjected to structured human review. E5 and E6 subsequently extended the same artefact to demonstrate subgroup-based evaluation and preprocessing variation, respectively.

3.5 Evaluation:

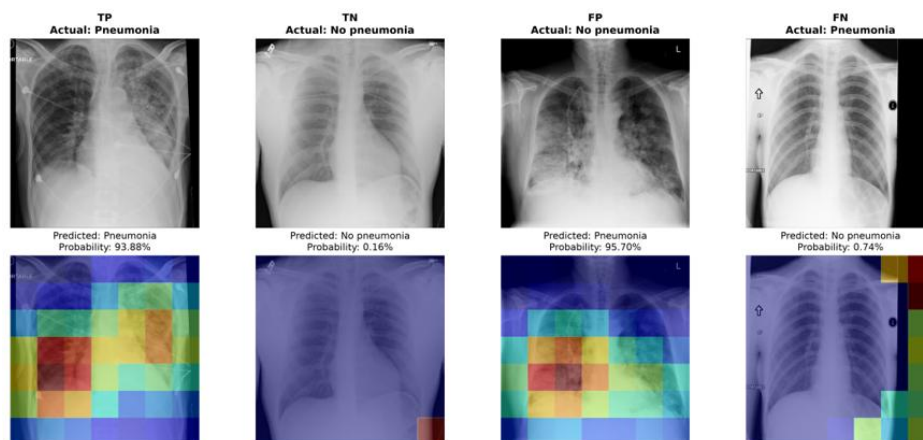
Evaluation was performed using the predefined patient-level dataset partitions, with the held-out test set reserved for final assessment. The held-out test set was used for final evaluation, with a probability threshold of 0.50 applied during training and validation and 0.75 used for test-set classification following validation threshold analysis.

E1 established the image-only DenseNet-121 baseline, while E2 incorporated age, sex, and radiographic view position. As shown in **Diagram**, E2 improved accuracy, precision, specificity, F1-score, and ROC-AUC, although recall and PR-AUC decreased slightly. This provided a direct comparison of image-only and multi-modal diagnostic performance.



E3 assessed explainability using representative TP, TN, FP, and FN cases. Grad-CAM heatmaps and activation measures demonstrated differing spatial patterns across predictions, as illustrated in **Figure 3.X**. This evaluated the artefact's explanation capability rather than the clinical validity of the explanations.

Figure 3.X: Grad-CAM visualisations for representative true-positive, true-negative, false-positive, and false-negative cases



E4 assessed human oversight using the four representative cases. The reviewer agreed with two predictions and disagreed with two; both model errors were corrected, resulting in 100% reviewer accuracy for the selected cases. The evaluation therefore demonstrated the functionality of the human-in-the-loop component.

Figure 1Table 3.X: E4 human-in-the-loop evaluation

Evaluation measure	Result
Cases reviewed	4
Model accuracy	50%
Reviewer agreement with model	50%
Reviewer accuracy	100%
Model errors	2
Errors corrected	2 (100%)

E5 evaluated the E2 model across age, sex, and radiographic-view subgroups. As shown in **Table 3.X**, performance varied most noticeably between PA and AP views, while reduced sensitivity was observed in the 60+ age group. This demonstrated differences in model behaviour across subgroups within the held-out test set.

Table 3.X: E5 subgroup evaluation of the multi-modal model

Subgroup	n	Accuracy	Sensitivity	Specificity	F1	ROC-AUC
All test	9,326	0.9239	0.3026	0.9394	0.1627	0.7815
Age <40	3,205	0.9332	0.3086	0.9494	0.1894	0.7968
Age 40–59	4,071	0.9167	0.3438	0.9306	0.1630	0.7722
Age 60+	2,050	0.9234	0.2157	0.9415	0.1229	0.7819
Female	4,072	0.9197	0.2600	0.9363	0.1372	0.8034
Male	5,254	0.9271	0.3359	0.9419	0.1834	0.7642
PA	5,969	0.9648	0.1683	0.9785	0.1393	0.7123
AP	3,357	0.8511	0.4094	0.8684	0.1722	0.7884

Finally, E6 compared baseline and CLAHE preprocessing under matched conditions. **Table 3.X** shows that baseline preprocessing achieved the higher validation ROC-AUC (0.7982 versus 0.7846), indicating that CLAHE did not improve performance in this experiment. Collectively, the evaluations examined the artefact's predictive, explanatory, interactive, subgroup, and preprocessing characteristics.

Table 3.X: E6 preprocessing comparison

Metric	Baseline	CLAHE	Difference
Epoch 1 ROC-AUC	0.7300	0.7285	−0.0015
Epoch 2 ROC-AUC	0.7975	0.7680	−0.0295
Epoch 3 ROC-AUC	0.7982	0.7846	−0.0136
Epoch 3 F1-score	0.1175	0.1102	−0.0073
Epoch 3 Sensitivity	0.7452	0.7019	−0.0433
Epoch 3 Specificity	0.7536	0.7516	−0.0020

3.6Communication:

4. Research Design Decisions:

5. Reliability, Validity and Rigor:

6. Ethical Considerations:

7. Limitations of the Research Design:

8. Conclusion:

3.6 Communication

The final DSR activity concerns communicating the artefact, its design decisions, evaluation results, and limitations to the research and practitioner communities.

The artefact is communicated through this research design paper, the documented experimental pipeline, quantitative evaluation metrics, Grad-CAM visualisations, HITL feedback records, subgroup analysis results, and preprocessing comparison results.

The implementation produces structured CSV outputs containing experimental metrics and case-level results, supporting an auditable record of the evaluation process.

The findings communicate several contributions. First, patient metadata can be integrated with a DenseNet-121 CXR classifier within a lightweight multi-modal architecture. Second, Grad-CAM can be incorporated to expose spatial characteristics associated with model predictions. Third, human review can provide a mechanism for detecting and correcting model errors. Fourth, subgroup evaluation demonstrates that aggregate diagnostic performance can conceal differences across

demographic and imaging groups. Finally, preprocessing evaluation demonstrates that alternative image enhancement techniques should be empirically validated rather than assumed to improve performance.

The communication of these findings also identifies the boundaries of the artefact, particularly the lack of external validation, the limited HITL sample, the limited XAI case sample, and the absence of a large-scale clinical user study.

4. Research Design Decisions

Several methodological decisions were made to ensure that the research remained aligned with the DSR methodology and the objectives established in the previous papers.

Design Science Research as the Primary Methodology

DSR was selected because the primary contribution of the study is the **design and evaluation of an artefact**, rather than the observation of an existing phenomenon alone. The framework represents an engineered solution to the identified problem and can therefore be evaluated according to its functionality, performance, interpretability, and utility.

The six activities of Peffers et al. (2007) provide the structure through which the artefact was developed, demonstrated, evaluated, and communicated.

Experimental Evaluation

Experimental evaluation was selected as a supporting method within the DSR process. E1 and E2 provide a comparison between image-only and multi-modal classification. E3 and E4 extend the evaluation beyond predictive performance to interpretability and human involvement. E5 investigates subgroup behaviour, while E6 examines the effect of preprocessing choices.

This combination allows the artefact to be evaluated from multiple perspectives rather than through a single performance metric.

Transfer Learning

DenseNet-121 was selected as the underlying image architecture to utilise established deep visual representations without requiring the development of an entirely new convolutional architecture. Transfer learning also supports the objective of developing a lightweight research artefact within the available computational resources.

Multi-Modal Integration

Age, sex, and radiographic view were incorporated as metadata because these variables are available alongside the CXR image and provide contextual information that is not directly represented by image pixels.

The purpose was not to claim that these variables are inherently diagnostic, but to investigate whether contextual patient and acquisition information can contribute to model discrimination.

Grad-CAM

Grad-CAM was selected as the primary XAI technique because it provides spatially interpretable heatmaps that can be directly associated with CXR regions. It also has precedent in pneumonia research, including Arias-Londoño et al. (2020) and Sheu et al. (2023).

SHAP and LIME were not implemented in the final artefact. This was a deliberate scope decision rather than an indication that these techniques are unsuitable. Implementing multiple XAI techniques would have increased experimental complexity without necessarily improving the core DSR demonstration.

Human-in-the-Loop Design

HITL was incorporated because an automated diagnostic system should not necessarily operate as an autonomous replacement for clinical judgement. Instead, the framework demonstrates a decision-support approach in which humans can inspect, challenge, and correct model outputs.

Subgroup Evaluation

Subgroup evaluation was incorporated to investigate whether aggregate model performance was representative of different demographic and imaging groups.

Age, sex, and radiographic view were selected because these variables were already available within the NIH dataset and form part of the metadata used by the multi-modal model.

The subgroup analysis is therefore intended to identify performance variation rather than to claim comprehensive fairness validation.

Preprocessing Design

The preprocessing comparison was introduced to determine whether an alternative image enhancement strategy could improve diagnostic performance.

CLAHE was selected because it can enhance local image contrast and is commonly considered for medical imaging applications. However, the experiment was designed as an empirical comparison rather than assuming that contrast enhancement would improve classification.

The results ultimately demonstrated that the baseline preprocessing pipeline produced higher final ROC-AUC and PR-AUC than CLAHE under the tested conditions.

5. Reliability, Validity and Rigor

Several measures were applied to improve the reliability, validity, and methodological rigor of the research.

First, the E1 and E2 models used the same NIH ChestX-ray14 dataset structure and evaluation procedure, allowing the contribution of patient metadata to be examined under comparable conditions. The E2 model achieved a best validation ROC-AUC of 0.7982 and was subsequently evaluated on the held-out test set as part of E5.

Second, multiple diagnostic metrics were used rather than relying solely on accuracy. Accuracy, precision, recall, sensitivity, specificity, F1-score, ROC-AUC, and PR-AUC were recorded. This is

particularly important because the evaluation dataset is highly imbalanced, with only 228 positive cases among 9,326 test images.

Consequently, the overall accuracy of 92.39% must be interpreted alongside the much lower precision of 11.13% and recall of 30.26%. The use of multiple metrics reduces the likelihood that conclusions will be driven by a single potentially misleading measure.

Third, patient-level splitting was used during dataset preparation. Patients were assigned to training, validation, and test partitions rather than randomly splitting individual images. This reduces the risk that images from the same patient appear across different partitions and artificially inflate performance.

Fourth, the XAI evaluation deliberately included TP, TN, FP, and FN cases. This reduces the risk of presenting only successful predictions and allows model explanation behaviour to be examined for both correct and erroneous outcomes.

Fifth, E5 expanded the evaluation beyond aggregate test performance by analysing age, sex, and radiographic-view subgroups. This provides additional evidence regarding the consistency of model behaviour within the evaluated population.

The subgroup results demonstrate why this analysis is important. For example, ROC-AUC ranged from 0.7123 for PA-view images to 0.8034 for female patients, while recall ranged from 16.83% for PA images to 40.94% for AP images.

Sixth, E6 provided a controlled preprocessing comparison. Both preprocessing configurations used the same multi-modal architecture and dataset structure, while the preprocessing strategy was changed. This enables the observed difference to be interpreted primarily in relation to the preprocessing choice.

The final E6 results showed that baseline preprocessing achieved a validation ROC-AUC of 0.7982 compared with 0.7846 for CLAHE. PR-AUC showed the same direction, with 0.0789 for baseline preprocessing compared with 0.0738 for CLAHE.

Seventh, the framework records model predictions, probabilities, metadata, explanation metrics, reviewer decisions, and corrected labels in structured output files. This creates an auditable experimental record and supports reproducibility.

Finally, the use of DSR provides methodological structure by explicitly connecting the identified problem, solution objectives, artefact, demonstration, evaluation, and communication activities.

Despite these measures, the study does not provide statistical evidence of clinical validity. The experiments should therefore be interpreted as evaluation of a research artefact rather than clinical validation of a deployable diagnostic system.

6. Ethical Considerations

The use of medical imaging and patient metadata introduces important ethical considerations. Although the study uses an existing research dataset rather than collecting new patient data, medical images and demographic attributes require careful handling.

The artefact was therefore designed as a research and decision-support system rather than an autonomous clinical diagnostic system. The predictions generated by the model should not be interpreted as medical diagnoses or as a replacement for qualified healthcare professionals.

Patient Privacy

The framework uses the de-identified NIH ChestX-ray14 dataset and limited metadata consisting of age, sex, and radiographic view. No directly identifying patient information is incorporated into the proposed diagnostic workflow.

Patient identifiers are used only as dataset-level identifiers necessary for patient-level data partitioning and evaluation.

Algorithmic Bias and Subgroup Differences

Algorithmic bias represents an important consideration because model performance can vary according to demographic and acquisition characteristics.

The E5 subgroup evaluation demonstrated measurable performance differences across age, sex, and radiographic view. For example, PA images achieved a ROC-AUC of 0.7123, whereas AP images achieved 0.7884. Female patients achieved a ROC-AUC of 0.8034 compared with 0.7642 for male patients.

These differences do not by themselves establish algorithmic discrimination or clinical unfairness, but they demonstrate why aggregate performance should not be treated as sufficient evidence of reliable behaviour across all populations.

The subgroup analysis therefore serves as an initial investigation of performance variation rather than a comprehensive fairness assessment.

Human Oversight

The incorporation of HITL provides an additional ethical safeguard because the framework positions the model as an aid to human decision-making rather than an unquestionable authority.

In the demonstrated cases, the reviewer was able to identify and correct both incorrect model predictions. However, the small sample and single-reviewer design prevent conclusions about clinical effectiveness.

Explainability

XAI introduces another ethical consideration. A Grad-CAM heatmap indicates image regions associated with a model prediction, but it does not establish that those regions represent a clinically verified pathological cause.

Consequently, Grad-CAM explanations should be used to support human inspection and model auditing rather than being treated as definitive clinical evidence.

Responsible Use

The framework should therefore be regarded as an experimental research artefact. Deployment in a clinical environment would require additional validation, appropriate regulatory assessment, security controls, clinical oversight, and evaluation by qualified medical professionals.

The results obtained in this study do not establish that the model is suitable for autonomous diagnosis or clinical deployment.

7. Limitations of the Research Design

Several limitations constrain the conclusions that can be drawn from this study.

First, the evaluation was conducted using the NIH ChestX-ray14 dataset. Although the E5 subgroup analysis evaluated differences across age, sex, and radiographic view, it remains a **within-dataset generalisability evaluation** rather than external validation. The study therefore does not establish whether the framework would perform similarly on images from other hospitals, imaging equipment, acquisition protocols, or patient populations.

Second, the positive class is substantially underrepresented. The E5 test set contained 228 positive cases compared with 9,098 negative cases. This imbalance contributes to the large difference between accuracy and positive-class metrics. Although the E2 model achieved 92.39% test accuracy, its precision was only 11.13% and recall was 30.26%.

Therefore, the high accuracy should not be interpreted as evidence of clinical-level pneumonia detection capability.

Third, the E5 subgroup analysis demonstrates that performance varies across different groups. In particular, PA-view images produced a ROC-AUC of 0.7123 and recall of only 16.83%, whereas AP-view images produced a ROC-AUC of 0.7884 and recall of 40.94%. These differences indicate that aggregate performance can conceal important variations in model behaviour.

However, the subgroup analysis is limited to the metadata variables available in the selected dataset. It does not constitute a comprehensive fairness or bias assessment across all clinically relevant demographic and socioeconomic characteristics.

Fourth, the XAI evaluation was intentionally limited to four representative cases. Although the inclusion of TP, TN, FP, and FN cases provides qualitative coverage of different prediction outcomes, four cases are insufficient to establish the statistical reliability or generalisability of explanation behaviour.

Fifth, only Grad-CAM was implemented. Although SHAP, LIME, Grad-CAM++, and other explanation methods have been investigated in previous research, the current artefact does not provide a systematic comparison of different XAI techniques.

Sixth, the HITL evaluation involved only four reviewed cases and a single reviewer. Although both model errors in the selected cases were successfully corrected, the result cannot be generalised to clinical professionals or larger populations. The 100% reviewer correctness observed in the selected cases should therefore not be interpreted as evidence of clinical reliability or trust.

Seventh, E6 was limited to a comparison between the baseline preprocessing pipeline and CLAHE. Although the experiment found that baseline preprocessing produced a higher final validation ROC-

AUC and PR-AUC than CLAHE, this does not establish that the baseline pipeline is universally optimal. Other preprocessing methods, image normalisation strategies, resolution choices, or augmentation techniques may produce different results.

Eighth, the E6 experiments were conducted using three training epochs because of the available computational resources. Although three epochs were sufficient to observe measurable differences between the preprocessing configurations, longer training could potentially alter the relative performance of the two approaches.

Ninth, the study does not constitute a clinical validation study. No prospective hospital trial, statistically meaningful clinician study, or comparison against radiologist performance was conducted.

Finally, the framework prioritised feasibility and demonstration within the available computational and time constraints. More extensive hyperparameter optimisation, external validation, larger-scale XAI evaluation, multiple XAI techniques, larger HITL studies, and additional demographic subgroup analysis could provide stronger evidence of effectiveness and generalisability.

8. Conclusion

This study designed and evaluated a lightweight multi-modal XAI deep learning framework for pneumonia diagnosis using Design Science Research. The resulting artefact integrates DenseNet-121-based image classification, patient metadata, Grad-CAM explanations, Human-in-the-Loop review, subgroup analysis, and preprocessing evaluation into a unified diagnostic decision-support workflow.

The E1 and E2 experiments established the progression from an image-only diagnostic model to a multi-modal architecture incorporating age, sex, and radiographic view. The E2 model achieved a best validation ROC-AUC of **0.7982**, demonstrating its ability to learn meaningful discriminatory patterns during model development.

Evaluation on the held-out NIH test set in E5 produced an overall accuracy of **92.39%**, specificity of **93.94%**, F1-score of **16.27%**, ROC-AUC of **0.7815**, and PR-AUC of **0.0858**. However, recall was only **30.26%** and precision was **11.13%**, demonstrating the limitations of relying on accuracy in a highly imbalanced diagnostic problem.

The E5 subgroup analysis further demonstrated that aggregate performance does not fully represent model behaviour. ROC-AUC ranged from **0.7123 for PA-view images** to **0.8034 for female patients**, while recall ranged from **16.83% for PA images** to **40.94% for AP images**. These findings demonstrate measurable variation across patient and imaging subgroups and support the inclusion of subgroup analysis when evaluating AI-based diagnostic systems.

The E3 XAI experiment demonstrated that Grad-CAM can expose differences in model activation between correct and incorrect predictions. The false-positive and false-negative cases demonstrated that prediction confidence and activation behaviour do not necessarily correspond directly to diagnostic correctness. This reinforces the value of examining model behaviour rather than relying exclusively on prediction probabilities.

The E4 HITL experiment further demonstrated that human review can provide a mechanism for identifying and correcting erroneous predictions. Both model errors in the selected four-case demonstration were successfully corrected by the reviewer. Nevertheless, the small sample size and single-reviewer design mean that these results should be interpreted only as a proof of concept rather than evidence of clinical trust.

Finally, E6 demonstrated that preprocessing choices can influence diagnostic performance. The baseline preprocessing strategy achieved a validation ROC-AUC of **0.7982** at the third epoch compared with **0.7846** for CLAHE. PR-AUC showed a similar pattern, with **0.0789** for baseline preprocessing compared with **0.0738** for CLAHE. Under the evaluated conditions, therefore, CLAHE did not provide an improvement over the baseline preprocessing pipeline.

Overall, the study demonstrates the feasibility of integrating **multi-modal learning, explainable AI, human oversight, subgroup evaluation, and preprocessing analysis within a single lightweight diagnostic framework.**

The primary contribution of the research is therefore not the achievement of state-of-the-art pneumonia classification accuracy. Instead, the contribution lies in the design and evaluation of an integrated DSR artefact that addresses several limitations identified in existing pneumonia CAD research.

The findings also demonstrate that evaluating an AI diagnostic framework requires more than a single aggregate accuracy measure. Diagnostic discrimination, positive-class detection, explanation behaviour, human correction, subgroup variation, and preprocessing sensitivity all provide important information about the behaviour and limitations of the artefact.

Future research should therefore focus on **external validation using independent datasets, larger clinician-based HITL studies, systematic comparison of XAI techniques including SHAP and LIME, broader subgroup and fairness analysis, optimisation of preprocessing and training strategies, and evaluation of explanation reliability.** Such work would provide stronger evidence regarding whether the proposed framework could ultimately support reliable clinical decision-making across diverse healthcare environments.

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